

Dot Situation

COMPLETE LEARNING GUIDE

What You'll Learn

Dot-situation questions show a base figure with 2 or 3 overlapping shapes (commonly a circle, a triangle, and a square) and a single dot placed somewhere inside the overlapping regions, then ask which of several answer options — each redrawing the same shapes in a different overlap arrangement — has a dot placed in the region satisfying the same inside/outside relationship to every shape. The reliable method is to convert the dot's position into an inside/outside code for each shape individually, since that code (not the dot's visual position on the page) is what must be matched across differently-arranged options. This sub-module builds that region-code method from a single base figure through the full 3-shape case, a Venn-diagram mental model, and the simpler verbal-condition variant where the code is given directly in words.

1. Reading the Base Figure's Dot Position

A dot-situation question typically shows a base figure with 2 or 3 overlapping shapes and a single dot placed somewhere within the overlapping regions — the first step is to determine, for EACH shape individually, whether the dot lies inside or outside that shape's boundary, ignoring the other shapes for the moment.

- A base figure has a circle, a triangle, and a square all overlapping. The dot sits in a region that is inside the circle's boundary, inside the triangle's boundary, but outside the square's boundary entirely. Checked one shape at a time: circle = inside, triangle = inside, square = outside.

2. Translating the Dot's Position Into a Region Code

Convert the inside/outside check for each shape into a compact code (such as circle: Yes, triangle: Yes, square: No) — this code is the actual answer requirement, completely independent of how the shapes happen to be arranged or overlap in the drawing, and it's what must be matched in the answer options, not the dot's visual position on the page.

- From the previous example (circle: inside, triangle: inside, square: outside), the region code is (C+, T+, S") — meaning 'inside circle AND inside triangle AND outside square.' Any candidate figure with a dot satisfying exactly this code is correct, regardless of where that region physically appears in that candidate's drawing.

3. Why the Answer Options Look Different From the Base Figure

Each answer option redraws the same 3 shapes but in a different relative arrangement or overlap pattern than the base figure — this is intentional, and it means you cannot simply look for 'the same visual spot' in each option; you must re-derive the inside/outside status for each shape fresh, in each option, using the region code as your only guide.

- The base figure has the triangle mostly overlapping the circle's left side. Option B instead draws the triangle overlapping the circle's right side, with the square positioned differently too. Despite the different layout, Option B is still evaluated by the same rule: find the region, if any, that is inside the circle, inside the triangle, and outside the square — matching the code, not the picture's layout.

4. Systematic Region Mapping for Three Overlapping Shapes

Three overlapping shapes create up to 7 distinct internal regions (each shape alone, each pair's exclusive overlap, and the region common to all three) plus the space outside all shapes — methodically identify all regions present in a given figure by checking each one against all 3 shapes before deciding which single region matches the required code.

- In a given answer option, checking the region where the triangle and square overlap (but the circle does not reach) gives the code (C" , T+ , S+) — this does NOT match the required (C+ , T+ , S") code from the earlier example, so this region is rejected even though it's a valid overlap region in the figure.

5. Worked Example: Matching a Code Across Multiple Options

Apply the full method end to end: state the required code from the base figure, then check each option's available regions against that code, eliminating options that have no region satisfying it and confirming the one option that does.

- Required code: (C+ , T+ , S"). Option A's regions include only (C+ , T+ , S+), (C" , T+ , S"), and (C+ , T" , S") — none match. Option B includes a region (C+ , T+ , S") exactly where its circle and triangle overlap outside the square's boundary — this matches, so Option B is the correct answer.

6. Pure Verbal-Condition Dot Situations

A variant skips the base figure entirely and instead states the condition directly in words (for example: 'in which figure does a dot lie inside the triangle and the square, but outside the circle') — treat this exactly like a pre-derived region code and search the options the same way, without needing a base figure to reverse-engineer the code from.

- Condition given in words: 'dot inside triangle and circle, outside square.' This is directly the code (Circle+ , Triangle+ , Square") — skip the base-figure derivation step entirely and go straight to checking each option's regions against this code.

7. Using a Venn-Diagram Mental Model

Thinking of the 3 shapes as a standard 3-circle Venn diagram (even when they're drawn as different shapes like a triangle and square) makes region identification far more reliable than trying to judge positions visually — mentally label regions using set logic (only-A, A-and-B-only, all-three, and so on) exactly as you would for a probability or set-theory Venn diagram.

- Treating circle=A, triangle=B, square=C, the required region (C+, T+, S") is exactly the Venn-diagram region 'A")B, excluding C' — the part of A and B's overlap that does not also fall inside C. Searching for this specific Venn region in each option is more reliable than trying to eyeball dot positions.

8. The Simpler Two-Shape Case

With only 2 overlapping shapes, there are just 4 possible regions (inside both, inside only the first, inside only the second, outside both) — this simpler case is worth solving explicitly first if it appears, since the same systematic logic scales up directly to the 3-shape case.

- Two overlapping shapes, a circle and a square. Required condition: dot inside the square but outside the circle. This is the region code (Circle" , Square+) — the crescent-shaped part of the square that does not overlap the circle. Any option whose square-only (non-overlapping) region contains the dot satisfies this.

Common Pitfalls

- Trying to match the dot's visual position on the page instead of deriving and matching its inside/outside code for each shape.
- Forgetting to re-derive the region code fresh for each answer option, since shapes are rearranged differently in each one.
- Checking only 2 of the 3 shapes' inside/outside status and missing a mismatch on the third shape.
- Confusing a pairwise overlap region (like circle-and-triangle-only) with the full three-way overlap region.
- In verbal-condition variants, misreading which shapes should be 'inside' versus 'outside' in the stated condition.
- Assuming a region that looks visually similar in size or position to the base figure's dot region is automatically the same code.

Key Takeaways

- Convert the dot's position into an inside/outside code for each shape individually — this code, not the visual position, is what must be matched.
- Each answer option rearranges the shapes differently; re-derive the code fresh in every option rather than pattern-matching the picture.

- Model 3 overlapping shapes as a Venn diagram with up to 7 distinct regions to identify overlaps systematically.
- Verbal-condition dot situations give the code directly in words — skip straight to matching regions in the options.
- Always check ALL shapes in the code (not just 2 of 3) before confirming a region match.
- Master the simpler 2-shape (4-region) case first; the 3-shape (7-region) case uses the identical logic.

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